LAB 5: SEQUENCE DIAGRAM - CHECKOUT & PAYMENT PROCESSING

Submitted by: Sparsh Tyagi, Vaishnavi Rajawat, Shubham Patel, Varnit Tomar **Faculty Mentor:** Mr. Rishabh Rathore Sir

AIM

To draw the Sequence Diagram for the customer checkout, payment verification, and automated multi-vendor sub-order creation scenario, modeling message exchange in strict temporal order.

THEORY & MESSAGE TRACE

A Sequence Diagram is an interaction diagram detailing how operations are executed over time. It highlights participant lifelines, activation bars, call messages, and return responses.

No.	From → To	Message / Operation Call	Description
1	Customer → :CartService	`checkout(cartId, addressId)`	Initiates checkout with saved delivery address.
2	:CartService → :InventoryManager	`validateAndLockStock(items)`	Verifies live stock quantities and reserves items.
3	:CartService → :PaymentGateway	`createTxToken(amount)`	Generates secure payment intent token for SDK.
4	:PaymentGateway → :OrderService	`webhook: paymentSuccess(txId)`	Cryptographically verified payment status payload.
5	:OrderService → :OrderSplitter	`splitOrderByVendor(cartItems)`	Groups cart items by `vendorId` into sub-orders.

SEQUENCE DIAGRAM

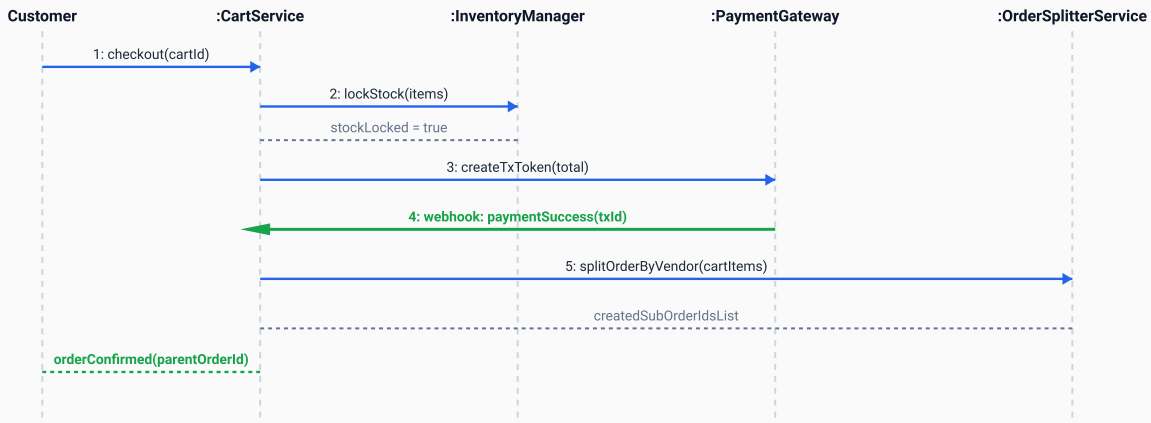


FIGURE 5.1: SEQUENCE DIAGRAM FOR CHECKOUT & MULTI-VENDOR ORDER SPLITTING

CONCLUSION

The Sequence Diagram models the exact time-ordered interaction required to execute stock checks, receive asynchronous payment webhooks, and trigger automatic order splitting.